

SILAS M. FALDE

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SUMMARY

Data science engineering student at the University of Michigan who builds reliable Python, C++, Bash, web, and Linux systems, from self-hosted infrastructure and automation to data-intensive applications.

EDUCATION

University of Michigan School of Engineering
Senior B.E.S. in Data Science, GPA: 3.09/4.00

Ann Arbor, MI
August 2023 – December 2026

Relevant Courses: Machine Learning, Artificial Intelligence, Computer Vision, Computational Modelling of Complex Systems, Database Management, Regression Analysis, Probability and Statistics, Bayesian Data Analysis, Data Structures and Algorithms

EXPERIENCE

Ally Financial, Inc.

Intern, Artificial Intelligence & Advanced Analytics

Detroit, MI
May 2026 – August 2026

- **Prototyped large language model workflows** to automate internal tasks and improve efficiency and safety.
- **Built Python NLP pipelines** to evaluate PII masking and redaction model performance for sensitive company documents.
- **Enhanced** a data dictionary migration tool by adding **batch processing**, smarter metadata selection, and a streamlined interface to improve usability and efficiency.

University of Michigan Department of Epidemiology

Research Assistant – Computational Lead

Ann Arbor, MI
September 2025 – May 2026

- Led development of a Python pipeline to classify collective-violence narratives using prompt-based large language model inference and structured response processing.
- Built reusable preprocessing utilities for NVDRS-style text, demographic derivation, path management, and annotation-round data handling.
- Designed human annotation workflows and measured inter-annotator agreement with Cohen's kappa and Krippendorff's alpha to resolve disagreements and assess model performance.

Ally Financial, Inc.

Intern, Data Management

Detroit, MI
May 2025 – August 2025

- **Created** data dictionaries and lineage documentation in **Python** to improve data governance and compliance.
- **Added data quality checks** to existing data pipelines with **Python and SQL** to improve data integrity and reliability.
- **Developed a Power BI** data availability dashboard to monitor refresh rates and availability for critical data assets.

PROJECTS

Home Server Infrastructure

Systems Developer

Ann Arbor, MI
2024 – Present

- **Designed and operate** a single-host **Ubuntu self-hosting platform** using **Docker Compose** stacks for ingress, media, cloud storage, AI services, monitoring, automation, and game hosting.
- **Automated media workflows** with **Bash and Python** for queueing, GPU-accelerated HEVC encoding, staging, metadata-safe publishing, library audits, and failure-aware batch processing.
- **Configured** Cloudflare Tunnel and Caddy ingress, Tailscale private administration, systemd scheduling, service health checks, and documented upgrade, backup, recovery, and security procedures.

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Web Developer

Ann Arbor, MI
2024 – Present

- **Built and maintain a responsive static personal website** with modular HTML, CSS, and JavaScript for resume, research, project, education, and photography pages.
- **Developed browser-side navigation, responsive layouts, photography galleries, asset loading, and reusable site-content modules** without a Node build system.
- **Automated resume publishing** by parsing LaTeX section files into browser-friendly JSON, and generated photography event manifests from organized image folders.

PyNVDRS

Package Developer

Ann Arbor, MI
2025 – 2026

- **Refactored reusable utilities** from NVDRS-style epidemiology projects into a typed, installable Python package with editable-install support.
- **Organized shared modules** for paths, narrative text cleaning, demographic preprocessing, annotation workflows, and optional GPT-assisted inference.
- **Added import and path tests, package metadata, optional dependency groups, and continuous integration** to support migration from research scripts.

Photo Processing Helper

Project Developer

Ann Arbor, MI
Jan 2026 – Present

- **Built a modular Python CLI and library** for photo framing, collage generation, panorama stitching, and Nikon RAW-to-JPG matching across large image folders.
- **Implemented OpenCV, NumPy, Pillow, and rawpy pipelines** for resizing, cropping, alignment, blending, and exact-dimension output while preserving high-bit-depth detail.
- **Added tests and validation** for file discovery, output integrity, stitching behavior, and repeatable batch processing.

Photo Library Compressor

Python Developer

Ann Arbor, MI
Personal Project

- **Engineered a Python batch pipeline** to compress and reorganize large iCloud photo libraries using file size, format, and GPS metadata.
- **Preserved EXIF metadata** while applying configurable quality steps, routing missing-location images, and tracking processing results in CSV metrics.
- **Added timeout handling, staged processing, and failure-tolerant orchestration** for long-running HEIC and JPEG media workloads.

Database Storage and Grace Hash Join

C++ Developer

Ann Arbor, MI
Course Project

- **Implemented C++ database-storage components** for disk pages, records, memory buffers, buckets, and relation management.
- **Built the partition phase of a Grace Hash Join** using page-level I/O, hash-based bucket placement, buffer flushing, and disk-page tracking.
- **Worked across a multi-file C++ codebase** with Make-based builds and explicit memory, page-capacity, and record-layout constraints.

Model-Based and Model-Free Reinforcement Learning

Reinforcement Learning Developer

Ann Arbor, MI
Course Project

- **Implemented Active ADP**, a model-based reinforcement-learning agent that learned transition and reward models and planned with value iteration.

- **Implemented Q-learning with optimistic exploration** and compared model-based and model-free learning in deterministic and stochastic Gymnasium FrozenLake environments.
- **Built evaluation, learning-curve, video-recording, and reproducibility utilities** with controlled random seeds and configurable experiments.

FPS PvP Dynamics
Project Developer

Ann Arbor, MI
March 2026 – April 2026

- **Built** a configurable **agent-based simulation** in Python to model player movement, combat, objectives, respawning, and stochastic per-tick match dynamics.
- **Developed trace capture, metrics export, visualization, and automated tests** for debugging and validating simulation behavior.
- **Ran parallel parameter sweeps and replication studies, Latin hypercube sampling, and Sobol sensitivity analysis** to quantify the effect of model parameters.

SKILLS

Programming: Python, C++, Bash, JavaScript, R, SQL

Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, spaCy, NLTK

Web: HTML, CSS, JavaScript, responsive design, static-site publishing

Systems: Linux, Docker Compose, systemd, CMake, Make, Git, CI/CD

Software: VS Code, Microsoft Office Suite, Google Workspace, Snowflake, Collibra, Alation, Manta